CS766 Project Proposal: Predicting Pedestrian Crossing Intention for Autonomous Vehicles

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1 Problem Statement

The primary challenge in urban autonomous driving is the high degree of uncertainty surrounding pedestrian behavior. Relying solely on observed movement leads to reactive safety measures that are often too late to prevent collisions. Moreover, predictive accuracy is frequently inconsistent across diverse demographic groups, introducing ethical and operational risks. This project proposes a proactive solution to this problem by modeling crossing intentions, while simultaneously analysing and mitigating demographic bias through logit masking and fairness-aware optimization.

2 Why is this problem important?

For years, autonomous systems have excelled at the ability to identify that a pedestrian exists. However, we are now at a critical turning point where the industry is moving beyond reactive safety toward proactive anticipation. A reactive system only brakes once a pedestrian has already stepped into the street, which is often too late to prevent a tragedy. This project explores intent-aware forecasting, investigating methods for vehicles to interpret environmental cues and anticipate pedestrian crossing behavior before they leave the curb.

This shift toward predictive modeling, however, brings a new set of responsibilities regarding the equity gap in AI safety. If a vision system is even slightly less accurate at predicting the movements of a child, a person using a wheelchair, or an elderly individual simply because they move differently than the "average" pedestrian in a training set, the technology has failed. Safety should not be a demographic privilege. Addressing this problem is vital because it ensures that as models become more predictive, they also become more equitable, using fairness-aware optimization to guarantee that every person on the street, regardless of age, gender, or mobility, is seen and protected with the same level of precision.

3 Current State-of-the-Art (SOTA)

The current landscape is defined by three major trends. First, efficient vision models like Efficient-PIE [1] have achieved real-time performance (up to 7.4 times faster than previous benchmarks) by using sole observation (single-frame) inference. The downside is that while these models are fast, they're a bit shallow. They might see that you're moving, but they don't really understand why or what you're planning to do next.

To fix that, Vision-Language Models (VLMs) like PedVLM [2] and OmniPredict utilize CLIP-based encoders and T5 language models to provide both intent prediction and explainable textual rationales for a vehicle's decision-making. The catch is that these models are often too slow to keep up with a car moving at city speeds, and they sometimes cheat by just looking at whether the car is already slowing down rather than actually paying attention to the person on the sidewalk.

Finally, Generative Diffusion Models are emerging as the SOTA for trajectory forecasting, as they can model multiple kinematically plausible future paths rather than a single deterministic "average" trajectory. These new models understand that people are unpredictable, as from one spot you might walk straight, turn left or stop to tie your shoe.

4 Proposed Approach

The project will begin by implementing current state-of-the-art (SOTA) architectures for pedestrian intent, specifically utilizing the PIE [3] and JAAD [4] datasets. For intent estimation, we plan to explore high-performance models such as EfficientPIE/PIP-Net [1, 5], which are optimized for real-time urban scenarios. The initial milestone will be the successful reproduction of the evaluated results reported in existing literature to establish a verified performance baseline.

Following this, the work will transition into a diagnostic phase to analyze inherent algorithmic biases. We will investigate "vehicle-side bias," where models may over-rely on the ego-vehicle's deceleration rather than the pedestrian's body language or appearance, and evaluate performance disparities across different demographic subgroups, such as children and the elderly.

To mitigate identified biases, we plan to experiment with techniques such as logit masking to reduce deterministic errors and improve model robustness. If time permits, we would also like to implement trajectory estimation and collision prediction logic, exploring methods like image-plane risk-aware mapping to assess the intersection of predicted pedestrian trajectories with the vehicle's path.

5 Evaluation and Timeline

To validate the project's impact within this tight window, we will focus on the following:

- **Baseline Performance:** Accuracy and F1-score on PIE [3]/JAAD [4] to ensure the baseline matches current benchmarks.
- **Fairness:** Using the demographic parity metric to see if the model is unsafe for specific groups (e.g., higher error rates for children).
- **Safety (if time permits):** Mean Average Displacement (ADE) and Final Displacement Error (FDE) for trajectory forecasting, alongside precision/recall for collision risk detection.

Table 1: Project Timeline and Milestones

Weeks	Milestones
Week 1–2	Data & Baseline: Preprocess PIE/JAAD into a unified format. Implement EfficientPIE/PIP-Net in Colab Pro. Match SOTA results.
Week 3–4	Analyze results by age and gender. Identify "Vehicle-Side Bias" (where model fails if the car doesn't slow down). Mid-term Report.
Week 5–6	Mitigation & Logic Masking: Apply Logit Masking to prioritize fairness. Fine-tune reasoning for vulnerable subgroups.
Week 7–8	If time permits: Collision prediction logic via trajectory estimation. Finalize visualizations, performance tables, and project webpage.

References

[1] Fang Qu, Pengzhan Zhou, Yuepeng He, Kaixin Gao, Youyu Luo, Xin Feng, Yu Liu, and Songtao Guo. 2025. EfficientPIE: real-time prediction on pedestrian crossing intention with sole observation. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI '25).*

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